| Question | Answer | Marks |
|---|---|---|
| 7(a) | the constant / direct current (in a resistor) | **M1** |
| | that causes the same <u>mean</u> power dissipation as the alternating current | **A1** |
| 7(b)(i) | mean power = **3.2 W** | **A1** |
| 7(b)(ii) | $P = I^2R$ | **C1** |
| | $I_{RMS} = \sqrt{(3.2 / 26)}$ or $[\sqrt{(6.4 / 26)}] / \sqrt{2}$<br>= **0.35 A** | **A1** |
| 7(c)(i) | diode | **B1** |
| 7(c)(ii) | <u>sketch:</u><br>*either*: line showing $P = 0$ between $t = 0$ and $t = 0.0125$ s and between $t = 0.0375$ s and $t = 0.050$ s and non-zero $P$ in between<br><br>*or*: line showing $P = 0$ between $t = 0.0125$ s and $t = 0.0375$ s, and non-zero P either side | **M1** |
| | line in non-zero $P$ regions with correct sine$^2$ shape, sitting on the $t$-axis, with maximum $P$ at 6.4 W | **A1** |
| 7(c)(iii) | mean power is now half of what it was before rectification | **B1** |
| | $I_{RMS} = \sqrt{(1.6 / 26)}$<br><br>= **0.25 A** | **A1** |